Shehla Amir

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Research Interests: Radio-over-Fiber, MIMO Signal Processing, Optical/Wireless Convergence, Joint Com

Education

North Carolina State University — **Raleigh, North Carolina**
PhD in Electrical Engineering — *Jan 2022 – Dec 2026 (expected)*
Advisor: Prof. Robert W. Heath Jr.
Relevant Courses: *Space-time communication theory, Signal processing and deep learning for advanced MIMO systems, LTE and 5G communications, Wireless Communications, Internet Protocols, Electromagnetic Fields, RF Design for wireless, Detection and Estimation theory, Photonics, Optical communications, Information theory*

North Carolina State University — **Raleigh, North Carolina**
MS in Electrical Engineering — *2026*

Lahore University of Management Sciences — **Lahore, Pakistan**
Bachelor of Science in Electrical Engineering — *Jul 2021*
Awards: Dean's Honor List 2018, 2019, 2020
Relevant Courses: *Electromagnetic Field and Waves, Communication systems, Computer Networks, Deep Learning, Digital Signal Processing, Principles of Optics, Stochastic systems, Advance Digital Signal Processing, Power Electronics*

Skills

Research: Wireless Communications, Radio-over-Fiber, MIMO Signal Processing, Optical/Wireless Convergence, Joint Communication & Radar, Information Theory
High-Level Languages: C, C++, Python
Algorithm Development: MATLAB
Design and Simulation Tools: LTspice, Proteus, Cadence AWR, HFSS, Quadriga, USRPs (SDR)
Technical Tools: MikroC, MPLAB, Adobe Photoshop, Wireshark

Experience

Graduate Research Assistant
North Carolina State University — *Jan 2022 – Present*
Supervisor: Prof. Robert W. Heath Jr.
- Combining optical communication theory with wireless communication for future radio access networks
- Using signal processing techniques to design efficient optical/wireless MIMO systems
- Prior work on joint communication and radar for multi-node radar systems

Research & Development Engineering Intern — **Apopka, FL**
Qorvo — *May 2023 – Aug 2023*
Supervisor: Dr. Mark Gallagher
- Understanding the working principles of surface acoustic waves for RF circuit design
- Automated the COM model vs. measurement analysis for surface acoustic wave (SAW) filters
- Streamlined the SAW filter design flow in Cadence AWR/HFSS using the pyawr library

Research Assistant — **Energy Storage Systems**
Lahore University of Management Sciences — *Jun 2021 – Nov 2021*
Supervisor: Dr. Ijaz Haider Naqvi
- Data-driven estimation of remaining useful life (RUL) of Li-ion batteries
- Equivalent circuit modeling to estimate battery state-of-health (SOH)

Teaching Experience

Teaching Assistant
North Carolina State University

Responsibilities: Supported instructor in grading exams and labs. Assisted students in performing the labs and held weekly office hours for any detailed help required.

1. ECE 200: Introduction to Signals, Circuits and Systems *(Summer 2026)*
2. ECE 211: Electric Circuits *(Spring 2025, Spring 2026)*
3. ECE 220: Analytical Foundations of ECE *(Fall 2024, Spring 2026)*
4. ECE 302: Microelectronics *(Fall 2025)*
5. ECE 331: Principles of Electrical Engineering *(Summer 2026)*

Teaching Assistant
Lahore University of Management Sciences

Responsibilities: Supported instructor in grading assignments, quizzes, labs and vivas. Assisted students in performing the lab experiments and held weekly office hours and tutorials for any detailed help required.

1. EE 220: Digital Logic Circuits *(Spring 2020, Spring 2021)*
2. EE 330: Electromagnetic Fields and Waves *(Fall 2020)*
3. EE 380: Communication Systems *(Fall 2020)*

Research Projects

MIMO aided by Radio-over-Multi-Mode Fiber (A-RoMMF) *Sept 2024 – Ongoing*
Supervisor: Prof. Robert W. Heath Jr. · with Dr. Miguel Rodrigo Castellanos
Motivation: Multi-mode fiber (MMF) spatially multiplexes signals over parallel optical modes, giving analog radio-over-fiber (A-RoF) a way to carry MIMO spatial streams over a single-wavelength fronthaul link without the added wavelength or per-core hardware overhead that wavelength-division-multiplexed single-mode fiber (SMF/WDM) requires. Since MMF mode-coupling strength changes how dispersion, nonlinear interference, and achievable capacity behave, this project develops closed-form models across both the strong- and weak-mode-coupling regimes so A-RoMMF performance can be predicted and compared to SMF/WDM without relying solely on measurement campaigns.
Work:

- Modeled the uplink A-RoMMF link as a cascaded wireless–optical MIMO channel that jointly captures wireless fading, fiber mode coupling, amplification (ASE) noise, and nonlinear interference, rather than treating the wireless and optical hops in isolation
- Under the strong-mode-coupling regime (SCR), compared MMF-aided A-RoF against WDM-aided A-RoF over SMF, showing MMF reduces nonlinear interference and supports higher data rates, limited mainly by the number of optical modes and wireless channels
- Under the weak-mode-coupling regime (WCR), derived a closed-form ergodic spectral efficiency (with low- and high-SNR approximations) and, via a Mellin-transform approach, the first closed-form outage probability for a cascaded wireless–MMF channel
- Characterized mode-dependent nonlinear interference covariance under WCR, showing A-RoMMF tolerates launch powers beyond 40 dBm before nonlinear penalties – well past the normal SMF operating region – while achieving spectral efficiency comparable to WDM/SMF
- Showed that for a representative configuration (12 receive antennas, 20 km analog fronthaul, 10 bits/s/Hz target rate) the cascaded channel preserves the full wireless diversity order, with the wireless hop – not the fiber – as the primary performance bottleneck

Related Publications:

C1 "Analog radio-over-multimode fiber: A spectral efficiency perspective," Proc. IEEE ICC, Montreal, QC, Canada, Jun. 2025 (strong mode-coupling regime)
J1 "Performance analysis of uplink analog radio-over-multimode fiber," submitted to IEEE Open J. Commun. Soc., 2026 (weak mode-coupling regime)

Cell-Free Massive MIMO Aided by Analog Radio-over-Multimode Fiber *2025 – Ongoing*

Supervisor: Prof. Robert W. Heath Jr.
Motivation: In a cell-free massive MIMO deployment, distributed remote radio heads (RRHs) jointly serve users and forward signals to a central unit over fronthaul links, so fronthaul architecture directly shapes achievable per-user performance. This project evaluates whether centralized analog radio-over-multimode-fiber (A-RoMMF) fronthaul – where each RRH connects to the central unit over multimode fiber – can approach the performance of an ideal, lossless fronthaul link in a multi-RRH, multi-user cell-free system.
Work:
- Modeled a cell-free MIMO network with multiple RRHs connected to a central unit over multimode fiber, combining A-RoMMF with WDM for the fronthaul
- Compared per-user uplink spectral efficiency against three baselines: ideal (perfect) fronthaul, WDM-aided A-RoF over single-mode fiber, and 1-bit (fully digitized) radio-over-fiber
- Showed centralized A-RoMMF fronthaul achieves per-user uplink spectral efficiency close to the ideal-fronthaul bound, and outperforms both WDM-SMF A-RoF and 1-bit RoF, which suffers a severe SE penalty from coarse quantization

Related Publication:
J3 "Cell-free massive MIMO aided by analog radio-over-multimode fiber," in preparation for IEEE Trans. Wireless Commun., 2026

A-RoF and Hybrid Beamforming — *Ongoing*
Supervisor: Prof. Robert W. Heath Jr.
Motivation: A WDM-based A-RoF fronthaul carries each precoded baseband stream on its own wavelength over single-mode fiber to the radio unit, where hybrid beamforming is applied. Because the optical path introduces its own wavelength-dependent phase shifts and temperature-sensitive chromatic dispersion, the fiber link itself can distort the array response used for beamforming. This project examines how much of that distortion actually degrades system performance, and designs a hybrid beamforming scheme that accounts for it.
Work:
- Modeled a WDM-based A-RoF fronthaul architecture, with baseband-precoded streams electrical-to-optical converted onto separate wavelength channels over single-mode fiber and recombined at the radio unit for hybrid beamforming
- Characterized the phase shift between WDM channels due to wavelength spacing and its effect on the array factor, finding that WDM phase shift alone does not significantly impact system performance
- Modeled the shift in fiber chromatic dispersion with temperature and its effect on the beamforming array pattern, showing that temperature changes in the fiber distort the array response and require periodic recalibration
- Proposed an ADMM-based hybrid beamforming design for A-RoF (A-RoF ADMM) and benchmarked its spectral efficiency across SNR against fully digital beamforming, standard ADMM hybrid beamforming, and an ideal (drift-free) RoF ADMM baseline
- Showed that temperature drift in the A-RoF fronthaul – not the WDM phase shift – is the dominant factor behind the spectral-efficiency gap versus fully digital beamforming

Power Consumption & Spectral Efficiency of Uplink Analog Radio-over-Fiber — *2024 – 2026*

Supervisor: Prof. Robert W. Heath Jr. · with Dr. Miguel Rodrigo Castellanos
Motivation: Radio-over-fiber centralizes radio access networks over a low-loss optical link between the remote radio head and the central unit. Analog radio-over-fiber (A-RoF) transmits RF signals directly on an optical carrier, avoiding digitization at the remote radio head and shifting power-hungry processing to the baseband unit. Prior comparisons of A-RoF against digital fronthaul relied on simplified, linearized optical models and did not account for fiber nonlinear interference or noise propagation – this work builds an analytical framework that does.
Work:
- Developed a general discrete-time input–output model for the combined optical-wireless uplink channel, capturing chromatic dispersion, electrical-to-optical conversion loss, amplification noise, and fiber nonlinear interference alongside wireless path loss and fading
- Derived the mutual information between the transmitted UE signal and the received BBU signal to compare A-RoF against DSP-assisted A-RoF and digital receivers, and to quantify the performance loss from quantization at low-resolution ADCs
- Modeled system power consumption – noise from the RF receiver, and power loss from E/O conversion and amplification – showing A-RoF achieves higher energy efficiency than 8- and 16-bit digital receivers and DSP-assisted A-RoF, despite being more sensitive to nonlinear interference
- Characterized trade-offs among transmit power, nonlinear interference, and spectral efficiency, identifying the linear operating regions where A-RoF is most effective for uplink wireless communication

Related Publication:
J2 S. Amir, M. R. Castellanos, and R. W. Heath, "Power consumption and spectral efficiency analysis for uplink analog radio-over-fiber," submitted to IEEE J. Lightw. Technol. [Online]. Available: `https://arxiv.org/abs/2606.01411`

Empowering Large Arrays from Electromagnetics and Optics — *Jan 2023 – Oct 2023*
Supervisor: Prof. Robert W. Heath Jr.
Motivation: Explored more efficient radio-over-fiber solutions that reduce complexity and power consumption while improving scalability, and worked to enhance digital RoF performance by optimizing signal processing and MIMO models for higher data rates and reliability.
Work:
- Reviewed alternative analog radio-over-fiber solutions to existing CPRI/eCPRI architecture
- Reviewed signal processing algorithms and MIMO models for digital radio-over-fiber architectures
- Presented findings at BWAC meetings, Spring and Fall 2023

Distributed Joint Communication and Radar Utilizing 5G Waveform — *May 2022 – Jul 2023*
Supervisor: Prof. Robert W. Heath Jr.
Motivation: Adaptive awareness of vehicles on the road enables more efficient design and execution of applications like cruise control. In a joint multi-node scenario, where vehicles communicate directly without relying on a central unit, each node becomes fully aware of its surroundings, enhancing overall system responsiveness and safety.
Work:
- Proposed a distributed vehicular JCR feedback model for target localization
- Developed a JCR transceiver that utilizes the 5G SSB for target detection and localization

Related Publication:
C2 "Multi-node joint communication and radar using synchronization signals in 5G," Proc. IEEE CAMSAP, Herradura, Costa Rica, Dec. 2023

Multiple Person Breathing Rate Estimation Using RFID — *Aug 2020 – May 2021*
Supervisors: Dr. Momin Ayub Uppal, Dr. Muhammad Tahir
Motivation: Contactless and non-invasive breath monitoring applications are becoming necessary. Given the pandemic, the need increased tenfold. Utilizing commercial off-the-shelf RFID readers, we can achieve this goal.
Work:
- Used Impinj's low-level reader protocol toolkit to extract raw phase and RSSI values
- Created different datasets for a user in different environments, such as stationary or moving users, and multiple users
- Removed phase jumps and noise from the phase data to extract a clean breathing signal to estimate the breathing rate

Dynamic Equivalent Circuit Modeling for Li-ion Battery State-of-Health — *Jan 2020 – May 2021*

Supervisors: Dr. Ijaz Haider Naqvi, Prof. Nauman Ahmad Zaffar
Motivation: As Electric Vehicles gain popularity, the demand for efficient battery management systems (BMS) is increasing. To build a reliable BMS we need to effectively quantify the degradation due to ageing and remaining useful life.
Work:
- Modeled lithium-ion batteries using an equivalent circuit model fit to the provided dataset
- Used curve-fitting tools to approximate the State-of-Health (SOH) of a lithium-ion battery
- Developed a novel 2-RC equivalent circuit model for SOH estimation with improved accuracy and lower computational cost than existing models; extended to multi-RC modeling, showing the 2-RC model achieves a global minimum

Related Publication:
- S. Amir, M. Gulzar, M. O. Tarar, I. H. Naqvi, N. A. Zaffar, and M. G. Pecht, "Dynamic Equivalent Circuit Model to Estimate State-of-Health of Lithium-Ion Batteries," IEEE Access, vol. 10, pp. 18279–18288, 2022

Side Projects

UAV Tracking Using 5G-V2X Joint Communication and Radar — *Aug 2023 – Dec 2023*
Motivation: 3GPP introduced 5G-V2X to improve road safety and traffic management in Intelligent Transportation Systems (ITS). In Ultra-Dense Networks (UDNs), tracking vehicular user equipment (VUE) is crucial, especially in dense urban environments where Roadside Units (RSUs) ensure constant line-of-sight, allowing accurate tracking even without GPS, enhanced by 5G's MIMO and beamforming capabilities.
Work:
- Developed simulations for localizing moving targets using an extended Kalman filter
- Designed the simulation environment with distributed MIMO stations following the METIS channel model

Maximally Flat Microstrip Bandpass Filter Design Using AWR — *Aug 2022 – Dec 2022*
Motivation: The goal of this project is to design a 10 GHz maximally flat Butterworth bandpass filter using microstrip technology with a 3 dB bandwidth of 40 MHz and 50 Ω source/load resistance. The design uses lossy elements, tuned to minimize insertion loss, with coupled microstrips to achieve optimal performance.
Work:
- Designed a basic lumped circuit element and then modeled it into transmission line stubs to microstrip elements

Target Localization Using Joint Communication and Sensing (IEEE 802.11ad) — *Jan 2022 – May 2022*
Work:
- Investigated the IEEE 802.11ad waveform for sensing stationary and moving targets at a fixed distance

Unsupervised Abnormality Detection in Intelligent, Heterogeneous Autonomous Systems
Supervisor: Dr. Zubair Khalid
Motivation: Automatic anomaly detection in autonomous systems is a key challenge that can open different negotiation between autonomous systems in the future.
Work:
- Analysis and pre-processing of time series ROS Bag Dataset
- GP Regression model to predict the time series data and detect any anomaly present

Kitchen Recycling Robot
Supervisor: Dr. Mohammad Jahangir Ikram
Motivation: Designed a path-following robot for the National Engineering and Robotics Competition that simulates real-life kitchen recycling. The contest revolves around the development of an Arduino-processor-driven robot capable of pattern reading and boundary detection.
Work:
- Created a design layout for the robot arm, PCB, and feeding station
- Interfaced sensors – IMU, IR, and ultrasound – to Arduino
- Collected sensor data and implemented a feedback loop that kept the robot following a set path

Tools, Framework & Hardware: Arduino, C/C++, Proteus
Awards/Recognition: Won the Best Engineering Design Award at the end of the event

Microcontroller and Interfacing Lab Project

Microcontroller Based Laser Engraver
- Designed a microcontroller-based engraver
- The device takes a monochromatic image as input and, based on pixel information, engraves the print onto a wooden surface using a laser

Tools, Framework & Hardware: MikroC, MPLAB, Proteus

Publications

Journal

J1 **S. Amir**, M. R. Castellanos, and R. W. Heath, "Performance analysis of uplink analog radio-over-multimode fiber," submitted to *IEEE Open J. Commun. Soc.*, 2026.

J2 **S. Amir**, M. R. Castellanos, and R. W. Heath, "Power consumption and spectral efficiency analysis for uplink analog radio-over-fiber," submitted to *IEEE J. Lightw. Technol.* [Online]. Available: `https://arxiv.org/abs/2606.01411`

J3 **S. Amir** and R. W. Heath, "Cell-free massive MIMO aided by analog radio-over-multimode fiber," in preparation for submission to *IEEE Trans. Wireless Commun.*, 2026.

Conference Proceedings

C1 **S. Amir**, M. R. Castellanos, Y.-H. Nam, and R. W. Heath, "Analog radio-over-multimode fiber: A spectral efficiency perspective," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Montreal, QC, Canada, Jun. 2025.

C2 **S. Amir**, M. R. Castellanos, and R. W. Heath, "Multi-node joint communication and radar using synchronization signals in 5G," in *Proc. IEEE Int. Workshop Comput. Adv. Multi-Sensor Adaptive Process. (CAMSAP)*, Herradura, Costa Rica, Dec. 2023.

Other

- **S. Amir**, M. Gulzar, M. O. Tarar, I. H. Naqvi, N. A. Zaffar, and M. G. Pecht, "Dynamic Equivalent Circuit Model to Estimate State-of-Health of Lithium-Ion Batteries," *IEEE Access*, vol. 10, pp. 18279–18288, 2022.

Certifications

Natural Language Processing with Classification & Vector Spaces by DeepLearning.ai

Extra Curricular & Leadership Role

- **Vice President Training and Development - Sports at LUMS**
- **Vice Captain - LUMS Netball Team**
- **General Secretary - Alpha Society LUMS**
- **Represented LUMS in 19th National Netball Championship**
- **Director Media and Promotions - LUMS Sports Fest 2020**